

HAGEN GRESS

Ph.D. in Mechanical Engineering, Boston University, 2025

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PROJECTS

MICROELECTROMECHANICAL SYSTEMS (MEMS) RESONATORS

November 2021 - present | Boston University

- assembled and calibrated a Michelson interferometer to optically measure nanoscale displacements
- designed and fabricated 3D-printed MEMS resonators with integrated electrothermal actuation and piezoresistive detection circuits
- designed and built a water-proof test chamber to study the resonance behavior of MEMS resonators under constant fluid flow
- studied the thermomechanical noise of resonators at 0.3 – 70 MHz in viscous fluids
- built a phase-locked loop to track resonance frequencies in gas sensing applications
- mentored an undergraduate student in integrating a Peltier element for thermoelectric cooling into the test chamber

ANTIBACTERIAL SUSCEPTIBILITY TESTING

September 2019 - October 2021 | Boston University

- prepared a variety of bacterial cell cultures
- designed and fabricated PDMS-based microchannels with integrated electrodes
- measured low-frequency (< 10 Hz) electrical resistance fluctuations in bacterial suspensions using a 4-wire configuration and a lock-in amplifier
- studied the frequency noise of bacterial suspensions exposed to different antibiotics
- adapted the platform to run similar experiments on human breast cancer cells
- wrote a research proposal for NSF Research Experiences for Undergraduates

WORK EXPERIENCE

FRESENIUS MEDICAL CARE NORTH AMERICA | Intern

May 2023 – August 2023 | Lawrence, MA

- created a CAD assembly of a water filter for home hemodialysis equipment
- assembled test fixtures to measure the flow rates of peristaltic pumps

MILTENYI BIOTEC GMBH | Plastics Engineer

September 2018 – July 2019 | Bergisch Gladbach, Germany

- designed injection-molded components for a cell sorting device using CAD software
- created 3D-models and technical drawings for cell strainers

FRAUNHOFER INSTITUTE FOR LASER TECHNOLOGY (ILT) | Research Assistant

January 2017 – March 2018 | Aachen, Germany

- assessed the suitability of different conductive photopolymers for 3D-printing
- converted a droplet dispenser into a 3D-printer for hydrogels

BARTELS MIKROTECHNIK GMBH | Intern

December 2015 – July 2016 | Dortmund, Germany

- assembled and tested microfluidic pumps in quality control
- designed electrode arrays and 3D-printed microfluidic structures to create a digital droplet system for optical switch actuation via electrowetting-on-dielectric (EWOD)

SKILLS

MODELING

SolidWorks (CSWA-MD) • Solid Edge • PTC Creo • COMSOL

SOFTWARE

OriginLab • MATLAB • LaTeX • Photoshop • Illustrator

FABRICATION

Wet Bench • Plasma Oven • Spin Coater • Mask Writer • Mask Aligner • 3D-Printer • Thin Film Evaporator • SEM • Ball Bonder • PDMS Casting

TESTING EQUIPMENT

Oscilloscope • Preamplifier • Lock-in Amplifier • Function Generator • PID Controller • Network/Spectrum Analyzer • Filters, Attenuators, Bias Tees, Power Splitters

EDUCATION

BOSTON UNIVERSITY

PH.D. IN MECHANICAL ENGINEERING
Boston, MA
Cum. GPA: 3.9 / 4.0

RWTH AACHEN UNIVERSITY

M.Sc. IN MECHANICAL ENGINEERING
Aachen, Germany
Cum. GPA: 4.0 / 4.0

RWTH AACHEN UNIVERSITY

B.Sc. IN MECHANICAL ENGINEERING
Aachen, Germany
Cum. GPA: 3.2 / 4.0

REFERENCE

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